

# 04

/ 13 LECTURES

## IP Addresses and Subnetting — Part II

Solving real subnetting problems, VLSM allocation, IPv6, the default gateway decision, broadcast types, and an introduction to NAT.

Subnetting Problems

VLSM

IPv6 & Compression

Default Gateway

Broadcast Address

Network Address

NAT Intro

Port Forwarding

### WHAT THIS LECTURE GIVES YOU

From theory to doing: a step-by-step subnetting recipe with a lookup table, a full VLSM allocation for four departments, IPv6 with its compression rules, exactly how your device decides "direct or gateway?", the two broadcast types, and how NAT lets a whole house share one public IP — with worked numericals in exam format.

STUDY-READY EDITION

CONCEPTS · DIAGRAMS · NUMERICALS · EXAM PREP

## LEARNING OBJECTIVES — AFTER THIS LECTURE YOU CAN

- Solve subnetting problems **from host requirements**
- Apply **VLSM** to allocate IP space without overlap
- Understand **IPv6** format and compress addresses correctly
- Explain the **default gateway** decision (direct vs via gateway)
- Describe **NAT**: the problem it solves and how the translation table works

### Subnetting II

#### Recipe

hosts+2 → smallest  $2^N$

prefix = 32 - N

#### VLSM

different prefixes per subnet

allocate **LARGEST** first

#### IPv6

128-bit colon-hex

:: once only

no broadcast, no NAT

#### Gateway

dest AND mask = my net?

yes → direct · no → gateway

#### Broadcasts

limited 255.255.255.255

direct (subnet) x.x.x.255

#### NAT

(private IP:port) ↔ (public IP:port)

SNAT · DNAT · PAT

## 1 Subnetting — From Requirements to Subnet

### 1 Determine hosts needed

How many usable devices per subnet?

### 2 Find smallest $2^N \geq (\text{hosts} + 2)$

The +2 reserves network & broadcast addresses.

### 3 Host bits = N → Prefix = 32 - N

Block size =  $2^N$  · Usable =  $2^N - 2$ .

### 4 Slice the parent block at block-size boundaries

Network addresses jump in steps of the block size.

| Hosts needed | Min $2^N$ | Prefix | Usable hosts | Hosts needed | Min $2^N$ | Prefix | Usable hosts |
|--------------|-----------|--------|--------------|--------------|-----------|--------|--------------|
| 2            | 4         | /30    | 2            | 31–62        | 64        | /26    | 62           |
| 3–6          | 8         | /29    | 6            | 63–126       | 128       | /25    | 126          |
| 7–14         | 16        | /28    | 14           | 127–254      | 256       | /24    | 254          |
| 15–30        | 32        | /27    | 30           | 255–510      | 512       | /23    | 510          |

### Example 1 — need 50 hosts per subnet inside 10.0.0.0/24

```
50 + 2 = 52 → next power of 2 = 64 = 26 → host bits 6 → prefix /26 → block 64
10.0.0.0/26    hosts .1 – .62    broadcast .63
10.0.0.64/26   hosts .65 – .126   broadcast .127
10.0.0.128/26  hosts .129 – .190  broadcast .191
10.0.0.192/26  hosts .193 – .254  broadcast .255
```

### Example 2 — how many /28 subnets fit in a /24?

```
subnets = 2(new – old) = 2(28–24) = 16 subnets · hosts each = 24–2 = 14
```

## 2 VLSM — Variable Length Subnet Masking

VLSM lets each subnet have a **different prefix length** — matching each department's real need instead of wasting addresses. **Golden rule: allocate the largest subnet first**, then work downward.

### Allocate 192.168.10.0/24 for four departments

| Dept        | Hosts needed | +2 → 2 <sup>N</sup> | Prefix | Block |
|-------------|--------------|---------------------|--------|-------|
| Engineering | 60           | 62 → 64             | /26    | 64    |
| Marketing   | 30           | 32 → 32             | /27    | 32    |
| HR          | 10           | 12 → 16             | /28    | 16    |
| Management  | 5            | 7 → 8               | /29    | 8     |

Allocation — largest first, no overlaps:

```
Engineering  192.168.10.0/26  → .0 – .63    (62 usable)
Marketing     192.168.10.64/27 → .64 – .95    (30 usable)
HR            192.168.10.96/28 → .96 – .111   (14 usable)
Management   192.168.10.112/29 → .112 – .119  (6 usable)
Remaining:   .120 – .255 kept for future growth
```

**EXAM** VLSM questions are graded on **order + no overlap**. Always start at the parent network address, jump by each block size, and show the usable range and broadcast for every subnet.

## 3 IPv6 — the 128-bit Future (Already Here)

- **Why:** IPv4 exhausted — IANA ran out of free blocks in **2011**. IPv6 gives  $2^{128} \approx 3.4 \times 10^{38}$  addresses — effectively unlimited.
- **Format:** 128 bits written as **8 groups of 4 hex digits** separated by colons:  
2001:0db8:85a3:0000:0000:8a2e:0370:7334

#### Compression Rule 1 — drop leading zeros

0db8 → db8 · 0000 → 0

#### Compression Rule 2 — one :: for the longest zero-run

2001:0db8:0000:0000:0000:8a2e:0370:7334  
→ 2001:db8::8a2e:370:7334

#### TRAP

:: may be used **only once** per address (two runs of zeros + :: would be ambiguous). If two runs exist, compress only the **longer** one (ties → the left one).

## IPv4 vs IPv6

| Feature         | IPv4               | IPv6                          |
|-----------------|--------------------|-------------------------------|
| Size            | 32 bits            | <b>128 bits</b>               |
| Format          | Dotted decimal     | Colon-hex                     |
| Total addresses | ≈ 4.3 billion      | $3.4 \times 10^{38}$          |
| Header          | Variable (20–60 B) | Fixed 40 B                    |
| Broadcast       | Yes                | <b>No — multicast instead</b> |
| NAT             | Necessary          | <b>Not needed</b>             |
| ARP             | Yes                | Replaced by NDP               |
| Fragmentation   | Routers + hosts    | Source host only              |

| Common IPv6 addresses | Meaning                                  |
|-----------------------|------------------------------------------|
| ::1                   | Loopback (like 127.0.0.1)                |
| ::                    | Unspecified / any                        |
| fe80::/10             | Link-local (auto-configured, not routed) |
| ff00::/8              | Multicast                                |
| 2000::/3              | Global unicast (public Internet)         |

## 4 Default Gateway — Direct or Forward?

The **default gateway** is the router's IP on your local network — the door to every other network. Your device makes a simple decision for every destination IP X:

1. Compute `X AND my_subnet_mask`
2. `Result == my_network_address?`
  - YES → X is on my subnet → deliver directly (ARP for its MAC)
  - NO → different network → send to default gateway (ARP for gateway's MAC)

**Figure 1.** The routing decision every host makes before touching the network.

Worked decision — my IP 192.168.1.5/24, gateway 192.168.1.1

```
To 192.168.1.20: .20 AND 255.255.255.0 = 192.168.1.0 == my network ✓ → direct
To 8.8.8.8:      8.8.8.8 AND 255.255.255.0 = 8.8.8.0 ≠ my network → gateway
```

CHECK YOUR GATEWAY

TERMINAL

```
# Linux      ip route show # "default via 192.168.1.1 dev eth0"
# Windows    route print  # or ipconfig → "Default Gateway"
# macOS      netstat -rn | grep default
```

5 Broadcast & Network Addresses — the Full Picture

| Address in subnet A.B.C.D/N | Calculation                     | Example (/24) |
|-----------------------------|---------------------------------|---------------|
| Network                     | All host bits = 0 (IP AND mask) | 192.168.1.0   |
| First host                  | Network + 1                     | 192.168.1.1   |
| Last host                   | Broadcast – 1                   | 192.168.1.254 |
| Broadcast                   | All host bits = 1               | 192.168.1.255 |

Limited broadcast

255.255.255.255 — every host on the local subnet; **never forwarded** by routers. Used by DHCP DISCOVER.

Directed broadcast

All host bits = 1 for a *specific* subnet (e.g. 192.168.1.255) — "everyone in 192.168.1.0/24".

6 NAT — Sharing One Public IP

**The problem:** your ISP gives you **one** public IP; your house has **many** devices. **NAT** (Network Address Translation) lets all of them share it, by rewriting IP:port pairs and remembering the mapping.

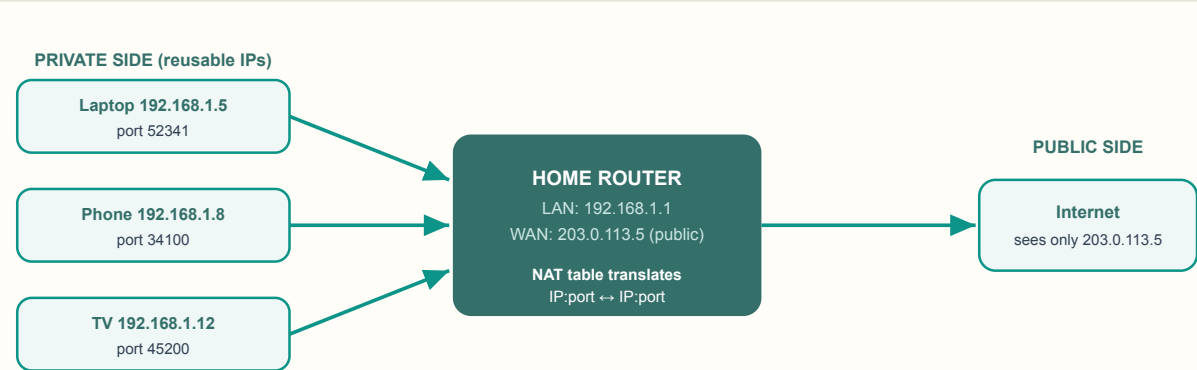


Figure 2. One public IP, many private devices — the router's NAT table keeps every conversation distinct by port.

### Outgoing (SNAT)

```
Laptop: 192.168.1.5:52341 →  
142.250.195.46:80  
Router rewrites src: 203.0.113.5:10001  
table: 10001 ↔ 192.168.1.5:52341
```

### Reply comes back

```
Google → 203.0.113.5:10001  
Router lookup 10001 → 192.168.1.5:52341  
Delivers to laptop ✓
```

| Variant                                   | Direction | Use case                                                                   |
|-------------------------------------------|-----------|----------------------------------------------------------------------------|
| SNAT (source NAT)                         | Outgoing  | Home internet sharing                                                      |
| DNAT (destination NAT)                    | Incoming  | <b>Port forwarding</b> , load balancing                                    |
| PAT (Port Address Translation / overload) | Both      | Many devices share one IP via distinct ports — what every home router does |

### Port forwarding (DNAT) — exposing a game server

```
Rule on router: public port 27015 → 192.168.1.10:27015  
Friend connects to 203.0.113.5:27015 → router DNATs → 192.168.1.10:27015
```

**COSTS** NAT **breaks end-to-end** (inbound needs port forwarding), is **stateful** (router tracks every flow; reboot kills sessions), and confuses protocols that embed IPs in their payload (FTP, VoIP/SIP, some games).

## 7 Worked Examples (Exam Pattern)

### Example 1 — 172.16.0.0/16 needs 500 subnets; hosts per subnet?

```
Host bits available = 32-16 = 16 · need  $2^N \geq 500 \rightarrow N = 9$  (512)  
Remaining host bits = 16-9 = 7 →  $2^7-2 = 126$  hosts/subnet · new prefix = /25
```

### Example 2 — compress 2001:0000:0000:0db8:0000:0000:0000:0001

```
Zero runs: "0000:0000" (groups 2-3) and "0000:0000:0000" (groups 5-7 — longer!)  
Compress the LONGER run: 2001:0:0:db8::1 (:: used once)
```

**SPEED TIP** Subnet-count ↔ host-count seesaw: every bit you take from hosts **doubles subnets and halves hosts-per-subnet**. Keep the seesaw in your head and most MCQs answer themselves.

## LECTURE 4 — RAPID REVISION CARD

**Recipe:** hosts+2 → smallest  $2^N$  → prefix = 32-N → block =  $2^N$ .

**Subnets** created =  $2^{(\text{new prefix} - \text{old prefix})}$ .

**VLSM:** allocate largest first; ranges must not overlap; show usable + broadcast.

**IPv6:** 128-bit, 8×4 hex groups; drop leading zeros; **::** **once** for the longest zero-run.

**::1** loopback · fe80::/10 link-local · 2000::/3 global · ff00::/8 multicast.

**Gateway rule:** dest AND mask == my network → direct; else → gateway.

**Limited broadcast** 255.255.255.255 (local only) vs **direct** x.x.x.255.

**NAT/PAT:** (priv IP:port) ↔ (pub IP:port) table; SNAT out, DNAT/port-forward in; breaks end-to-end.

## ✦ Important Questions — Exam Practice

### A. Multiple Choice

**Q1.** You need 100 usable hosts per subnet. The correct prefix is:

MCQ

- a) /25
- b) /26
- c) /27
- d) /24

**Q2.** How many /20 subnets can be carved from 10.0.0.0/16?

MCQ

- a) 8
- b) 16
- c) 4
- d) 32

**Q3.** The shortest form of 2001:0000:0000:0000:0001:0000:0000:0001 is:

MCQ

- a) 2001::1:0:0:1
- b) 2001:0:0:0:1::1
- c) 2001::1::1
- d) 2001:0:0:0:1:0:0:1

**Q4.** Your IP is 10.1.1.50/24 and you send to 10.1.2.5. The packet goes:

MCQ

- a) directly to 10.1.2.5
- b) to the default gateway
- c) to 255.255.255.255
- d) nowhere — invalid destination

**Q5.** A company receives a /22 block. Total usable host addresses:

MCQ

- a) 1024
- b) 1022
- c) 512
- d) 510

## B. Short Answer (2–4 marks)

**Q6.** State the 4-step subnetting recipe and apply it: 60 hosts per subnet in 192.168.50.0/24 — **SHORT**  
give the first three subnets with ranges.

**Q7.** What is VLSM and why does the "largest first" rule matter? **SHORT**

**Q8.** Differentiate limited and directed broadcast with addresses for 172.16.5.0/24. **SHORT**

**Q9.** What problem does NAT solve, and what problem does it create? (two each) **SHORT**

**Q10.** In NAT, who initiates connection tracking — the router or the device? Why does that matter for inbound connections? **SHORT**

## C. Long Answer / Numericals (5–8 marks)

**Q11.** Allocate 192.168.20.0/24 via VLSM for: Sales 100 hosts, Tech 50, Admin 20, Guest 10, WAN link 2. Show prefix, range, broadcast for each — largest first, no overlap. **NUMERICAL**

**Q12.** Explain how a packet from your laptop reaches 8.8.8.8 when your IP is 192.168.1.5/24 — include the gateway decision, ARP's role, and NAT rewriting (both directions). **LONG**

**Q13.** 172.16.0.0/16 must be split into 500 subnets. Find the new prefix, usable hosts per subnet, and the first two subnet ranges. **NUMERICAL**

**Q14.** Draw the NAT translation flow for two devices (laptop :52341 and phone :34100) talking to the same web server simultaneously. Why don't the replies get mixed up? **LONG**

### ANSWER KEY

**A1.**  $a - 100 + 2 = 102 \rightarrow 2^7 = 128 \rightarrow$  prefix /25 (126 usable).

**A2.**  $b - 2^{(20-16)} = 16$ .

**A3.**  $a$  — longest zero-run is groups 2–4; only one :: allowed.

**A4.**  $b - 10.1.2.0 \neq 10.1.1.0 \rightarrow$  off-subnet  $\rightarrow$  gateway.

**A5.**  $b - 2^{10} - 2 = 1022$ .

**A6.**  $60 + 2 = 62 \rightarrow 64 \rightarrow /26$ . Subnets: 192.168.50.0/26 (.1–.62, bc .63), .64/26 (.65–.126, bc .127), .128/26 (.129–.190, bc .191).

**A7.** Variable-length prefixes per subnet to match real need; largest first guarantees later smaller blocks fit in the remaining space without overlap.

**A8.** Limited 255.255.255.255 — local segment only, never routed. Directed 172.16.5.255 — everyone in that specific subnet.

**A9.** Solves: IPv4 scarcity, one public IP for many devices, hides internal topology (accidental firewall). Creates: breaks end-to-end inbound, stateful router, protocol breakage (FTP/VoIP).

**A10.** The router — it creates the entry when the **first outgoing** packet of a flow passes. That's why unsolicited inbound has no entry and is dropped unless port-forwarded.

**A11.** Sales .0/25 (1–126, bc 127) · Tech .128/26 (129–190, bc 191) · Admin .192/27 (193–222, bc 223) · Guest .224/28 (225–238, bc 239) · WAN .240/30 (241–242, bc 243). Remaining .244–.255 spare.

**A13.**  $2^N \geq 500 \rightarrow N=9 \rightarrow$  prefix /25; hosts =  $2^7 - 2 = 126$ ; first: 172.16.0.0/25 (.1–.126), second: 172.16.128.0/25 (.129–.254).

**A12/A14.** Use Figure 2 and the SNAT/reply cards; the key sentence: distinct source **ports** create distinct NAT entries, so reply packets lookup uniquely by port.